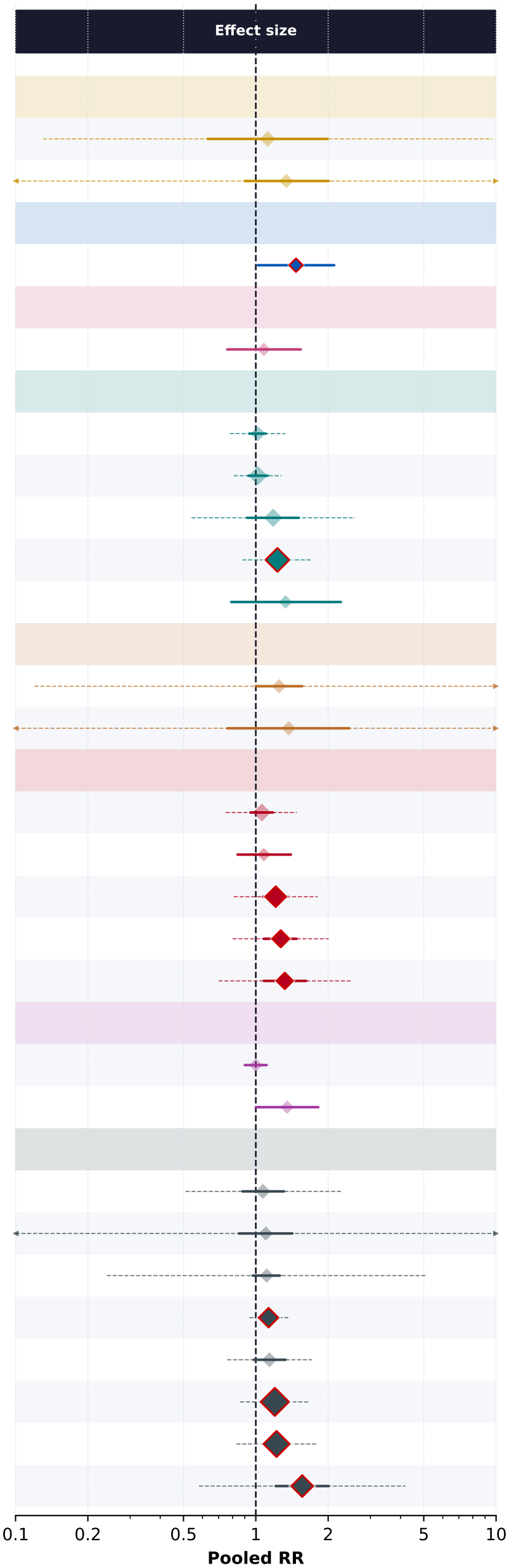
# Exposures positively associated with prostate cancer risk
## Meta-analysis of dietary-intake studies

| Exposure | Studies | N | Cases | Pooled RR (95% CI) |
|---|---|---|---|---|
| ***VITAMINS A, C, D, E, K*** | | | | |
| **Vitamin A** | 5 | 3,207 | 1,358 | *1.12 (0.63-1.99)* |
| **Vitamin D** | 3 | 247,791 | 29,373 | *1.34 (0.90-2.01)* |
| ***B VITAMINS*** | | | | |
| **Vitamin B12** | 2 | 719 | 245 | *1.47 (1.02-2.12)* |
| ***ANTIOXIDANTS*** | | | | |
| **Antioxidants** | 2 | 50,507 | 7,155 | *1.08 (0.76-1.54)* |
| ***MINERALS & TRACE ELEMENTS*** | | | | |
| **Iron** | 4 | 183,783 | 11,462 | *1.02 (0.94-1.10)* |
| **Selenium** | 10 | 235,441 | 18,652 | *1.02 (0.93-1.12)* |
| **Zinc** | 9 | 94,876 | 7,312 | *1.18 (0.92-1.51)* |
| **Calcium** | 19 | 770,289 | - | *1.23 (1.12-1.36)* |
| **Mineral supplements** | 2 | 57,251 | 1,958 | *1.33 (0.79-2.26)* |
| ***FERMENTED FOODS & PROBIOTICS*** | | | | |
| **Yogurt** | 3 | 201,324 | 7,680 | *1.25 (1.00-1.56)* |
| **Fermented foods** | 3 | 3,460 | 339 | *1.37 (0.76-2.45)* |
| ***FATTY ACIDS & LIPIDS*** | | | | |
| **Omega-3 fatty acids** | 8 | 607,190 | 20,690 | *1.06 (0.95-1.18)* |
| **Omega-9 fatty acids** | 2 | 200,518 | 7,678 | *1.08 (0.84-1.40)* |
| **Polyunsaturated Fat (PUFA)** | 13 | 673,771 | - | *1.21 (1.07-1.36)* |
| **Monounsaturated Fat (MUFA)** | 7 | 255,429 | - | *1.27 (1.08-1.48)* |
| **Alpha-linolenic acid** | 6 | 556,471 | 12,655 | *1.32 (1.08-1.62)* |
| ***DIET*** | | | | |
| **Mediterranean diet** | 2 | 106,146 | 7,476 | *1.00 (0.90-1.11)* |
| **Western pattern diet** | 2 | 44,462 | 1,702 | *1.35 (1.00-1.82)* |
| ***OTHER*** | | | | |
| **Eggs** | 4 | 928,359 | 53,580 | *1.07 (0.88-1.31)* |
| **Ultra-processed food intake** | 3 | 4,983 | 2,932 | *1.10 (0.85-1.42)* |
| **Whole-grain intake** | 3 | 79,458 | 7,209 | *1.11 (0.97-1.26)* |
| **Processed meat intake** | 11 | 1,119,262 | 65,902 | *1.13 (1.05-1.22)* |
| **Multivitamin** | 4 | 348,954 | 4,443 | *1.14 (0.98-1.33)* |
| **Alcohol** | 29 | 1,934,957 | - | *1.20 (1.10-1.30)* |
| **Dairy** | 24 | 739,672 | 33,998 | *1.22 (1.12-1.34)* |
| **Red meat** | 15 | 1,002,157 | 57,829 | *1.56 (1.21-2.01)* |




Exposure group

Vitamins A, C, D, E, K
B Vitamins
Antioxidants
Minerals & Trace Elements
Fermented Foods & Probiotics
Fatty Acids & Lipids
Diet
Other

*RR > 1 = positively associated with prostate cancer risk. | ◆ pooled RR (size reflects number of studies) | ◇ red outline = statistically significant (95% CI excludes 1.0) | ◆ faded = not statistically significant | dashed line = 95% prediction interval | arrowhead = interval continues beyond the axis*

Include subexposures: Yes